Torsion-Born Early Dark Energy with an Onset Locked to Matter–Radiation Equality: Derivation and Pre-Registered Likelihood Tests

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Abstract

We construct early dark energy (EDE) from spacetime torsion and test it against current data with a full Boltzmann code. Minimal Einstein–Cartan torsion cannot shorten the sound horizon: its spin–torsion contact energy is negative and 10^{-58}–10^{-57} of the radiation density at $z \simeq 3500$. Promoting the Barbero–Immirzi parameter γ to a field gives a canonical, ghost-free pseudoscalar $\phi = \sqrt{3/2}\, M_{\rm Pl}\, \mathrm{arsinh}(1/\gamma)$ whose kinetic energy is the torsion energy and whose screening is dynamical. With one assumed potential, $V = \Lambda^4[1 - \cos((\phi - \phi_0)/F)]^3$, the Lagrangian is that of axion-like EDE. We then (i) propose a reduced model, the *locked onset*, in which the field's critical scale factor is fixed to matter–radiation equality, $a_c = \omega_r/(\omega_b + \omega_c)$, removing one free parameter, and (ii) fit all variants with AxiCLASS and Cobaya to Planck 2018, DESI DR2 BAO and Pantheon+ (no distance-ladder prior), with every decision rule written down and hashed before each run. Relative to ΛCDM the free model improves the fit by $\Delta\chi^2 = -6.4$ (ΔAIC $= -0.4$, a tie) and its preferred onset ($z_c \simeq 3575$) sits next to equality ($z_{\rm eq} \simeq 3553$). The locked model fits equally well with two extra parameters, $\Delta\chi^2 = -6.4$, ΔAIC $= -2.4$, $H_0 = 70.6$, $f_{\rm EDE} = 0.078$; because the lock was chosen after the free fit, this is not a blind result. Re-minimising ΛCDM and the locked model with identical optimiser settings lowers the Planck 2018 gain to $\Delta\chi^2 = -4.70$ (ΔAIC $= -0.70$), robust to three other supernova compilations. On the pre-registered blind test with ACT DR6 (P-ACT + DESI DR2 + Pantheon+) the locked model improves the fit by $\Delta\chi^2 = -7.23$ (ΔAIC $= -3.23$, $\approx 2.2\sigma$, $H_0 = 70.4$), passing its rule. Fixing the width as well, to $F = \sqrt{3/2}\, M_{\rm Pl}/9$, leaves one free parameter and gives $\Delta\chi^2 = -6.14$ (ΔAIC $= -4.14$, $p = 0.013$, $\approx 2.5\sigma$), although a free fit of the divisor gives $N = 8.0$, missing the pre-registered 9 ± 0.5. Adding the DES Y3 + KiDS-1000 S_8 measurement reduces the Planck gain to $\Delta\chi^2 = -2.92$, so the pre-registered overall verdict is a tie. The ACT collaboration's own analysis finds no significant preference for early dark energy; our $\approx$ 2–2.5σ results are consistent with that. Fixing the width to the pure torsion value $F = \sqrt{3/2}\, M_{\rm Pl}$ is excluded ($\Delta\chi^2 = +14.6$); on Planck PR4 the free model gives no likelihood improvement; curvature is consistent with zero. The minimal model predicts no measurable cosmic birefringence, and its present-day spin coupling lies 15–19 orders of magnitude below laboratory bounds. We state plainly which parts of the construction are derived, which are assumed, and which results failed.

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1 Introduction

1.1 The problem

Distance-ladder measurements give $H_0 \simeq 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [4]. The Planck CMB analysis under flat ΛCDM gives 67.36 ± 0.54 [5]. The CMB measures the angle

$$\theta_* = \frac{r_s^*}{D_C(z_*)} \tag{1}$$

to about 0.03%. Here r_s^* is the comoving sound horizon at photon decoupling and $D_C(z_*)$ the comoving distance to it. If new physics shortens r_s^*, a shorter $D_C(z_*)$, and therefore a larger H_0, becomes compatible with the same angle. A component of energy that briefly matters near matter–radiation equality and then dilutes faster than radiation does exactly this. That is the idea behind early dark energy (EDE) [6, 7].

1.2 What this paper adds

EDE is usually introduced as a scalar field with no stated origin. Here we derive such a field from the geometry of spacetime, specifically from torsion, and show that the derivation fixes its sign, its kinetic structure and its screening behaviour. The paper is organised so that every step falls into one of three classes:

- **Derived:** follows from the stated action with no further input (§2–§3, §6).
- **Assumed:** the scalar potential $V(\phi)$ (§4). This is the only free functional input.
- **Computed:** every number, by code that is supplied (Appendix B).
- **Tested:** a reduced model whose onset is locked to matter–radiation equality (§5), and every variant fitted with a full Boltzmann code and likelihoods (§10–§11), with decision rules fixed and hashed in advance.

§14 collects the objections a referee is likely to raise and answers each one, or states that it remains open.

Conventions. Signature $(-,+,+,+)$; $\kappa = 8\pi G = M_{\rm Pl}^{-2}$ with $M_{\rm Pl}$ the reduced Planck mass; $\hbar = c = 1$ unless restored. Densities $\omega_i \equiv \Omega_i h^2$.

2 Einstein–Cartan gravity

2.1 Action and the Cartan equation

EC theory keeps the metric $g_{\mu\nu}$ but lets the connection be non-symmetric. Its antisymmetric part is the torsion,

$$T^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\mu\nu} - \Gamma^{\lambda}{}_{\nu\mu}. \tag{2}$$

In first-order (tetrad) form, with tetrad $e^a{}_\mu$ and spin connection $\omega^{ab}{}_\mu$ varied independently, the action including the Holst term is [9]

$$S_{\rm grav} = \frac{1}{2\kappa}\int d^4x\; e\; e^{\mu}{}_a e^{\nu}{}_b \left(R^{ab}{}_{\mu\nu} - \frac{1}{2\gamma}\,\epsilon^{ab}{}_{cd} R^{cd}{}_{\mu\nu} \right), \tag{3}$$

where $R^{ab}{}_{\mu\nu}(\omega)$ is the curvature of the spin connection and γ is the Barbero–Immirzi parameter [10, 11]. Overall signs of the Holst term differ between references; none of the results below depend on that choice. For *constant* γ and no fermions, the Holst term does not change the field equations. It matters only when γ varies, or when fermions are present.

Varying $\omega^{ab}{}_\mu$ gives the Cartan equation, which is algebraic in torsion. With minimally coupled matter of canonical spin tensor $s_{\mu\nu}{}^{\lambda}$ and constant γ it reads [13]

$$T^{\lambda}{}_{\mu\nu} + \delta^{\lambda}_{\mu} T^{\sigma}{}_{\nu\sigma} - \delta^{\lambda}_{\nu} T^{\sigma}{}_{\mu\sigma} = \kappa\, s_{\mu\nu}{}^{\lambda}. \tag{4}$$

Torsion does not propagate. It is fixed pointwise by its source and can be eliminated, leaving Einstein's equations with extra terms quadratic in the source. Those terms carry one extra power of κ.

2.2 Minimal EC with fermions: the null result

For a Weyssenhoff spin fluid with randomly oriented spins, eliminating torsion and averaging gives an effective perfect fluid with [15, 16]

$$\tilde{\rho} = \rho - \frac{\kappa}{4}\langle S^2\rangle, \qquad \tilde{p} = p - \frac{\kappa}{4}\langle S^2\rangle, \qquad H^2 = \frac{\kappa}{3}\left(\rho - \frac{\kappa}{4}\langle S^2\rangle\right). \tag{5}$$

Two treatments of fermionic matter appear in the literature, and they differ in their coefficient but agree in sign. For number density n (with $\hbar$ and c restored):

- *Spin fluid* (Hehl–von der Heyde–Kerlick averaging, $\langle S^2\rangle = \hbar^2 n^2/8$): $\tilde{\rho}_s = \tilde{p}_s = -\alpha_{\rm sf} n^2/c^2$ with $\alpha_{\rm sf} = \kappa(\hbar c)^2/32 = 6.49 \times 10^{-96}$ J m^3. Since $n \propto a^{-3}$, this term scales as a^{-6} with $w = 1$, which is consistent with $\dot{\rho}_s + 3H(\rho_s + p_s) = 0$.
- *Dirac fields* (totally antisymmetric spin tensor, the Hehl–Datta interaction [14]): Popławski [18] finds $\tilde{\epsilon}_s = -\alpha_D n^2$ and $\tilde{p}_s = +\alpha_D n^2$, with $\alpha_D = \frac{9}{16}\kappa(\hbar c)^2 = 1.17\times10^{-94}$ J m^3, eighteen times $\alpha_{\rm sf}$.

In both treatments the contribution to the energy density, and hence to H^2, is negative and scales as $n^2 \propto a^{-6}$.

The cosmic fermion number is dominated by relic neutrinos, $n_\nu = 3 \cdot \frac{3}{11} n_\gamma = 3.36 \times 10^8$ m^{-3} today. Electrons and nucleons are smaller by nine orders of magnitude. Then

$$\frac{\Omega_s}{\Omega_r}\Big|_{z=3540} = \begin{cases} -1.3 \times 10^{-58} & \text{(spin fluid)} \\ -2.3 \times 10^{-57} & \text{(Dirac)} \end{cases} \tag{6}$$

$|\rho_s|$ reaches ρ_r only near $T \sim 10^{19}$ GeV (7×10^{19} and 2×10^{19} GeV for the two coefficients), the regime of EC bounce cosmology [17, 18]. Minimal EC torsion therefore fails on two counts.

1. *Sign.* The contact energy is negative. It *lengthens* r_s^* and lowers the inferred H_0. A component with this scaling and the EC sign, of size $f = 0.05$ at $z_c = 3540$, gives $H_0 = 62.8$ at fixed θ_* (Appendix A).
2. *Magnitude.* The amplitude is fixed by Gn^2 and is 56–57 orders of magnitude too small. No coupling constant of order one can repair this.

Both failures share one cause: torsion sourced by spin is tied to particle density. The next section removes that tie.

3 The Barbero–Immirzi field

3.1 Promoting γ to a field

Let $\gamma \to \gamma(x)$ in (3), and write $\bar{\gamma} \equiv 1/\gamma$. The Holst term is no longer topological, and the Cartan equation acquires a source proportional to $\partial_\mu \bar{\gamma}$. Taveras and Yunes [12] solved it for vacuum gravity (matter not coupled to the connection). The torsion is algebraically determined by $\partial_\mu \bar{\gamma}$, and vanishes wherever $\bar{\gamma}$ is constant (their Eq. 15). Substituting the solution back into the action gives (their Eq. 49)

$$S_{\rm eff} = \frac{1}{2\kappa}\int d^4x\sqrt{-g}\left[R(g) - \frac{3}{2}\,\frac{g^{\mu\nu}\partial_\mu\bar{\gamma}\,\partial_\nu\bar{\gamma}}{1+\bar{\gamma}^2}\right], \tag{7}$$

with $R(g)$ the ordinary Levi-Civita curvature. All second derivatives of $\bar{\gamma}$ cancel identically.

3.2 Canonical field

Define ϕ by

$$\frac{1}{2\kappa}\cdot\frac{3}{2}\,\frac{(\partial\bar{\gamma})^2}{1+\bar{\gamma}^2}\equiv\frac{1}{2}(\partial\phi)^2 \quad\Longleftrightarrow\quad \frac{d\phi}{d\bar{\gamma}}=\sqrt{\frac{3}{2\kappa}}\,\frac{1}{\sqrt{1+\bar{\gamma}^2}}. \tag{8}$$

Since $\int d\bar{\gamma}/\sqrt{1+\bar{\gamma}^2}=\operatorname{arsinh}\bar{\gamma}$,

$$\phi=\sqrt{\tfrac{3}{2}}\,M_{\rm Pl}\,\operatorname{arsinh}\!\left(\frac{1}{\gamma}\right),\qquad \gamma=\frac{1}{\sinh\!\big(\phi/\sqrt{3/2}\,M_{\rm Pl}\big)}, \tag{9}$$

and (7) becomes

$$S_{\rm eff}=\int d^4x\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R-\frac{1}{2}(\partial\phi)^2\right]. \tag{10}$$

The map (9) is a monotonic bijection $\mathbb{R}\to\mathbb{R}$, so no field range is lost.

3.3 Three consequences that are derived, not assumed

1. **Positive energy, no ghost.** In signature $(-,+,+,+)$ the kinetic term $-\frac{1}{2}(\partial\phi)^2$ has the healthy sign. The energy density $\frac{1}{2}\dot{\phi}^2+V\geq 0$, and $\rho+p=\dot{\phi}^2\geq 0$, so the null energy condition holds at all times. This repairs the sign failure of §2.2.

2. **The kinetic energy is the torsion energy.** The kinetic term in (7) is precisely what remains after the torsion-squared terms are substituted out. Torsion scales as $|\partial\bar{\gamma}|\propto|\dot{\phi}|$, and the torsion contact energy is $\frac{1}{2}\dot{\phi}^2$.

3. **Built-in screening.** Torsion is non-zero only while ϕ moves. A field frozen by Hubble friction at early times has zero torsion. A field settled at a potential minimum at late times also has zero torsion. The screening profile that earlier drafts had to postulate is replaced by field dynamics.

The amplitude is no longer tied to Gn^2. It is set by the energy stored in ϕ, which repairs the magnitude failure of §2.2.

4 The one assumption: the potential

The effective action (10) has no potential. We add one, and this is the only element of the model not derived from (3):

$$V(\phi)=\Lambda^4\left[1-\cos\!\left(\frac{\phi}{F}\right)\right]^n. \tag{11}$$

Three considerations motivate this form. First, the Holst term is parity-odd, so $\bar{\gamma}$, and hence ϕ, is a pseudoscalar. Second, a pseudoscalar's shift symmetry is broken in the generic case by periodic, non-perturbative terms. Third, the power n controls how fast the energy dilutes after the field is released. Near the minimum, $V\simeq\Lambda^4(\phi/F)^{2n}/2^n$, and a field oscillating in a ϕ^{2n} potential has time-averaged equation of state [19]

$$w_n=\frac{n-1}{n+1}\qquad(n=1:0;\ n=2:\tfrac{1}{3};\ n=3:\tfrac{1}{2};\ n\to\infty:1). \tag{12}$$

For $n\geq 2$ the component dilutes at least as fast as radiation.

Where the field settles. Because only $\partial\phi$ and V' enter the dynamics, the potential's minimum can sit at any ϕ_0 without changing any cosmological prediction: write $V = \Lambda^4[1 - \cos((\phi - \phi_0)/F)]^n$. We place it at the value of the Immirzi parameter fixed by black-hole entropy counting in loop quantum gravity, $\gamma_0 \simeq 0.2375$ (Domagala–Lewandowski; Meissner), i.e. $\phi_0 = \sqrt{3/2}\, M_{\rm Pl} \operatorname{arsinh}(1/\gamma_0) \simeq 2.63\, M_{\rm Pl}$. The field then relaxes to the γ that loop quantum gravity requires today, and torsion still vanishes once ϕ stops moving. (Placing the minimum at $\phi = 0$, i.e. $\gamma \to \infty$, would switch the Holst term off entirely but contradict that entropy calibration.)

Parameters. $\{\Lambda, F, \theta_i, n\}$, with $\theta_i \equiv \phi_i/F$ the initial misalignment. We fix $\theta_i = 2.8$, close to the hilltop and in the range preferred by EDE fits [7], and report $n = 2, 3, 4$. Λ and F are then traded for the observable pair (z_c, f): the redshift at which the field's energy fraction peaks, and the peak fraction.

5 A reduced model: the locked onset

In the full-likelihood fits of §10 the EDE sector has three free parameters: the peak fraction $f_{\rm EDE}$, the critical redshift z_c (equivalently the scale factor a_c at which the field is released), and the initial angle θ_i. Each costs two units of AIC. When z_c is left free, its best-fit value lands next to matter–radiation equality computed for the same cosmology (§10). This motivates a reduced model in which the onset is not a parameter:

$$a_c \equiv a_{\rm eq} = \frac{\omega_r}{\omega_b + \omega_c}, \qquad \omega_r = \omega_\gamma \left(1 + \tfrac{7}{8}\left(\tfrac{4}{11}\right)^{4/3} N_{\rm eff}\right) = 2.4728 \times 10^{-5}\,(1 + 0.22711\, N_{\rm eff}), \quad (13)$$

with $N_{\rm eff} = 3.044$, evaluated for every trial cosmology from its own densities. The model has two extra parameters over ΛCDM ($f_{\rm EDE}$, θ_i).

Status of the lock. We do not derive Eq. (13). A physical reason to expect the field to be released when matter overtakes radiation could come from the fermion sector: the Holst action couples $\partial_\mu \bar{\gamma}$ to the fermion axial current (§15), so the field's environment changes character at equality. We have not computed this, and we present the lock as a phenomenological hypothesis that removes a tuned number and can be falsified: on any new data set the onset must coincide with equality. We also disclose that the lock was chosen *after* seeing the free fit; the first data set on which it is tested blind is that of §11.

The torsion width. If the potential were periodic with period 2π in the natural torsion variable $\operatorname{arsinh}(1/\gamma)$, the width would be $F = \sqrt{3/2}\, M_{\rm Pl} = 1.2247\, M_{\rm Pl}$. Combining this with Eq. (13) leaves a single free parameter. We test it in §10; it is excluded.

6 Cosmological dynamics

6.1 Equations in e-folds

Let $N = \ln a$ and $' \equiv d/dN$, so that $\dot{\phi} = H\phi'$. Write $H = 100\,\mathcal{E}$ km s^{-1} Mpc^{-1}, measure ϕ in units of $M_{\rm Pl}$, and measure energy densities in units of $3M_{\rm Pl}^2 H_{100}^2$, so that ω_i and $v \equiv V/(3M_{\rm Pl}^2 H_{100}^2)$ are dimensionless. Then:
Friedmann. $3M_{\rm Pl}^2 H^2 = \rho_{\rm bg} + \frac{1}{2}H^2\phi'^2 + V$ gives

$$\mathcal{E}^2 = \frac{\omega_m a^{-3} + \omega_r a^{-4} + \omega_\Lambda + v(\phi)}{1 - \phi'^2/6}. \quad (14)$$

Klein–Gordon. From $\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0$ and $\ddot{\phi} = H^2\phi'' + H\dot{H}/H\,\phi'$,

$$\phi'' + \left(3 + \frac{\mathcal{E}'}{\mathcal{E}}\right)\phi' + \frac{3\,v_{,\phi}}{\mathcal{E}^2} = 0. \tag{15}$$

Raychaudhuri. From $\dot{H} = -(\rho + p)/2M_{\rm Pl}^2$,

$$\frac{\mathcal{E}'}{\mathcal{E}} = -\frac{3}{2}\,\frac{\omega_m a^{-3} + \frac{4}{3}\omega_r a^{-4}}{\mathcal{E}^2} - \frac{\phi'^2}{2}. \tag{16}$$

The scalar energy fraction is $f_\phi(a) = \rho_\phi/\rho_{\rm tot} = 1 - (\omega_m a^{-3} + \omega_r a^{-4} + \omega_\Lambda)/\mathcal{E}^2$. We define $f \equiv \max_a f_\phi$ and z_c its location. Flatness gives $\omega_\Lambda = h^2 - \omega_m - \omega_r - \rho_\phi(1)$, where the last term is negligible.

6.2 Initial conditions and screening

Deep in radiation domination, $\phi = \phi_i$ and $\phi' = 0$. Hubble friction holds the field fixed while $H \gg m_{\rm eff}$, so $\rho_\phi = V(\phi_i)$ is constant and

$$f_\phi \propto a^4 \quad \text{(radiation era)}. \tag{17}$$

At $T = 1$ MeV this gives $f_\phi \approx 4 \times 10^{-25}$ for every model below, so BBN is unaffected. (This also corrects a claim in earlier drafts, that a profile of the form $2/[1+(a/a_c)^6]$ "does not screen". A constant early energy density *is* screened, because its fraction falls as a^4. That profile is the $n \to \infty$ fluid limit of (11) and is physically admissible.)

6.3 Observables

$$r_s(a_{\rm e}) = \frac{c}{H_{100}}\int_0^{a_{\rm e}} \frac{da}{a^2\mathcal{E}(a)\sqrt{3\big(1 + \frac{3\omega_b}{4\omega_\gamma}a\big)}}, \qquad D_C(z) = \frac{c}{H_{100}}\int_0^z \frac{dz'}{\mathcal{E}(z')}, \tag{18}$$

with $c/H_{100} = 2997.92458$ Mpc, $r_s^* = r_s(a_*)$ and $r_d = r_s(a_d)$. The decoupling and drag redshifts are taken from the Hu–Sugiyama [20] and Eisenstein–Hu [21] fits. The latter is rescaled by 1.03848 to reproduce Planck's $z_d = 1059.94$. Fixed inputs are $\omega_b = 0.02237$, $\omega_\gamma = 2.473 \times 10^{-5}$ and $N_{\rm eff} = 3.046$. The ΛCDM baseline ($\omega_c = 0.1200$, $h = 0.6736$) gives $r_s^* = 144.27$ Mpc, $r_d = 147.10$ Mpc, $\theta_* = 1.03834 \times 10^{-2}$ and $r_d h = 99.09$ Mpc. All fits below hold θ_* and $r_d h$ at these values.

7 Why Ω_m is pinned and ω_c must rise

This result is independent of the torsion sector, so it is worth deriving first.

Fixed ω_m fails. At fixed physical densities, h enters $D_C(z_*)$ only through $\omega_\Lambda = h^2 - \omega_m - \omega_r$. Differentiating,

$$\left.\frac{\partial \ln D_C(z_*)}{\partial \ln h}\right|_\omega = -h^2\,\frac{\int_0^{z_*}\mathcal{E}^{-3}dz}{\int_0^{z_*}\mathcal{E}^{-1}dz} = -0.1937. \tag{19}$$

Matching θ_* then amplifies any fractional cut in r_s^* about fivefold in h. This drives $\Omega_m = \omega_m/h^2$ and $r_d h$ far from their BAO values (Appendix A, Table 6).

Matching θ_* and $r_d h$ together. Write $\theta_* = (r_s^* h)/(h D_C)$. Across all solutions r_s^*/r_d changes by less than 0.03%, so fixing $r_d h$ fixes $r_s^* h$. Then θ_* fixes $h D_C(z_*) = (c/H_{100}) \int dz/E(z)$, which depends only on Ω_m (and weakly on Ω_r). Therefore Ω_m stays near its baseline value 0.314, and

$$\omega_c \simeq \Omega_m^{\rm base} h^2 - \omega_b \xrightarrow{h=0.728} 0.1439. \tag{20}$$

Any model that raises H_0 by shortening the sound horizon while respecting BAO must raise ω_c in proportion to h^2. The torsion field's role is to supply the reduction of r_s^* that makes this ω_c consistent with θ_*.

8 Background solutions

This section solves the background equations alone, holding θ_* and $r_d h$ at their ΛCDM values, to show what the model can do in principle. The full-likelihood fits, which also fit perturbations and all other parameters, are in §10; they prefer a smaller $f_{\rm EDE}$ and H_0. The system (14)–(16) is integrated from $a = 10^{-8}$ to 10^{-2} with an 8th-order Runge–Kutta scheme (relative tolerance 10^{-10}). Beyond $a = 10^{-2}$ the scalar density is continued with the fitted power law, and it is negligible. For each Lagrangian parameter set we solve for (h, ω_c) such that θ_* and $r_d h$ match the baseline. Both residuals are below 10^{-15} in every case.

8.1 The original target point

Tuning (Λ, F) so that $f = 0.12$ and $z_c = 3540$ at ΛCDM densities ($\Lambda^4 = 4.641 \times 10^8$ in units of $3M_{\rm Pl}^2 H_{100}^2$, $F = 0.1683\, M_{\rm Pl}$, $n = 3$), the joint fit gives

$$H_0 = 70.65, \quad \omega_c = 0.1343, \quad \Omega_m = 0.3139, \quad r_s^* = 137.56 \text{ Mpc}, \quad r_d = 140.25 \text{ Mpc}, \tag{21}$$

with (z_c, f) shifting slightly to $(3507, 0.119)$ at the fitted densities.

8.2 Models reaching $H_0 = 72.8$

Fixing $z_c = 3540$ and solving for the F that yields $h = 0.728$:

Four features of Table 1 stand out. (a) The measured late-time equation of state reproduces (12), which validates the integration. (b) ω_c agrees with the geometric prediction (20) to 0.3% for all n. (c) All field values are sub-Planckian. (d) The energy scale $\Lambda \approx 0.5$ eV is set by the requirement $m_{\rm eff} \sim H(z_c)$. It is not explained by the model (§14).

9 Relation to early dark energy and current data

After the field redefinition (9), the model's Lagrangian, (10) plus (11), is *identical* to axion-like EDE [7]. This has two consequences, and both should be stated openly.

- **Strength.** Every published Boltzmann-code and likelihood analysis of axion EDE, including perturbations, applies to this model without modification. The model is not an untested construction. It is a geometric origin for a well-studied one.
- **Limitation.** In the pure-gravity sector the model makes no prediction that distinguishes it from EDE. Distinguishing signatures require the fermion sector (§15).

The most recent combined analysis, of Planck, ACT DR6 and DESI DR2 BAO by Poulin, Smith, Calderón and Simon [8], finds the following.

- Without the distance-ladder prior, the 95% upper limit is $f(z_c) < 0.109$, with $H_0 = 69.71^{+0.64}_{-1.3}$.

	$n = 2$	$n = 3$	$n = 4$
w_n predicted, Eq. (12)	0.333	0.500	0.600
w measured over $z = 100$–1000	0.334	0.495	0.555
$F/M_{\rm Pl}$	0.1496	0.2139	0.2767
$\phi_i/M_{\rm Pl} = \theta_i F$	0.419	0.599	0.775
γ_i from Eq. (9)	2.87	1.97	1.48
Λ [eV]	0.566	0.510	0.451
peak fraction f	0.159	0.183	0.201
f_ϕ at z_*	0.059	0.044	0.022
f_ϕ at 1 MeV	3.5×10^{-25}	4.5×10^{-25}	5.3×10^{-25}
H_0 [km s^{-1}Mpc^{-1}]	72.80	72.80	72.80
ω_c	0.1437	0.1441	0.1442
Ω_m	0.3133	0.3141	0.3143
r_s^* [Mpc]	133.50	133.48	133.46
r_d [Mpc]	136.11	136.11	136.11

Table 1: Torsion-born EDE solutions with $z_c = 3540$, $\theta_i = 2.8$, and θ_*, $r_d h$ held at the ΛCDM baseline. For $n = 4$ the late-time w is still approaching its asymptotic value at $z = 100$.

- With it, $f(z_c) = 0.126 \pm 0.021$ and $H_0 = 72.34 \pm 0.72$.
- The profile likelihood without the distance ladder gives $f = 0.090^{+0.030}_{-0.033}$ and $H_0 = 71.96^{+1.08}_{-0.99}$, a preference for $f > 0$ at about 2.5σ.

The residual tension with the distance ladder falls below 2σ, but the authors note a worsened Ω_m tension with Pantheon+ and DES-Y5 supernovae.

Our background-only estimate requires $f \simeq 0.18$ ($n = 3$) for $H_0 = 72.8$. That is larger than the full-likelihood values above, for two reasons. We hold $r_d h$ at the Planck ΛCDM value, which is lower than DESI prefers. We also keep n_s and the other primordial parameters fixed, although they shift in a full fit. The data-preferred range, $f \approx 0.09$–0.13 with $H_0 \approx 72$, is the region a full analysis should target. In plain terms, current data allow this model to reduce the tension substantially, not to eliminate it. Our own full-likelihood fits (§10) find $f_{\rm EDE} = 0.078$ and $H_0 = 70.6$ without the distance-ladder prior.

10 Full-likelihood fits

10.1 Pipeline and data

We use AxiCLASS [22, 23], the CLASS extension that evolves the field ϕ and its perturbations for the potential (11) with $n = 3$, and Cobaya [24] for the likelihoods. Best fits are found with the derivative-free BOBYQA optimiser [25, 26], from several independent starting points, with a final "polish" restart from the best point. The baseline data are:

- Planck 2018 [27, 28]: low-ℓ TT and EE, high-ℓ Plik TT,TE,EE (`lite`), and CMB lensing;
- DESI DR2 BAO [29];
- Pantheon+ supernovae [30, 31], without the SH0ES distance-ladder prior.

We also use Planck PR4 (CamSpec high-ℓ [32] and PR4 lensing [33]) in place of the Planck 2018 high-ℓ and lensing likelihoods.

Fit statistic. The optimiser minimises the posterior, which includes the Gaussian prior on the Planck calibration $A_{\rm planck}$ (1 ± 0.0025). Two independent ΛCDM optima have the same posterior to 0.04 but likelihood-only χ^2 values that differ by 1.6, the difference being absorbed by the calibration prior. We therefore compare models with $\chi^2 \equiv \chi^2_{\rm like} + [(A_{\rm planck} - 1)/0.0025]^2$, which is the quantity consistently minimised, and quote likelihood-only values alongside. The ΛCDM reference is $\chi^2 = 2434.22$ ($\chi^2_{\rm like} = 2434.06$). Model comparison uses $\Delta{\rm AIC} = \Delta\chi^2 + 2k$ for k extra parameters [37].

Pre-registration. Before each test below we wrote the hypothesis, the data and the pass/fail rule into a text file and recorded its SHA-256 hash; the files are in the repository (Appendix B). The rule for the locked onset was $\Delta\chi^2 < -4$ against ΛCDM, i.e. $\Delta{\rm AIC} < 0$.

10.2 Results

Model	k	$\Delta\chi^2$	$\Delta\chi^2_{\rm like}$	ΔAIC	H_0	$f_{\rm EDE}$
ΛCDM	0	0	0	0	67.98	–
Free onset ($f_{\rm EDE}, z_c, \theta_i$)	3	−6.43	−6.52	−0.43	70.60	0.078
Locked onset, Eq. (13)	2	**−6.39**	−6.48	**−2.39**	70.60	0.078
Locked, $F = \sqrt{3/2}\, M_{\rm Pl}$	1	+14.63	+13.82	+16.63	71.19	0.074
ΛCDM, Ω_k free	1	−0.86	−0.93	+1.14	68.04	–
ΛCDM, $\Omega_k = -0.013$ fixed	0	+165.2	+163.3	+165.2	65.98	–
Locked onset, Ω_k free	3	−6.79	−7.00	−0.79	70.60	0.078

Table 2: Best fits to Planck 2018 + DESI DR2 + Pantheon+ (no SH0ES). $\Delta\chi^2$ includes the Planck calibration prior (see text); $\Delta\chi^2_{\rm like}$ is likelihood only. H_0 in ${\rm km\,s^{-1}Mpc^{-1}}$. These rows come from the original single-run fits and are superseded for the free and locked onsets by Table 3, whose numbers are converged and use one engine for every model.

Model	k	χ^2	$\Delta\chi^2$	ΔAIC	ΔBIC	H_0
ΛCDM (best of 8)	0	2433.58	0	0	0	68.33
wCDM	1	2433.22	−0.36	+1.64	+7.2	68.03
Free onset ($f_{\rm EDE}, z_c, \theta_i$)	3	2429.74	−3.84	+2.16	+18.9	69.65
Locked onset, Eq. (13) (best of 8)	2	2429.62	−3.96	+0.04	+11.2	69.93

Table 3: Same-engine comparison (CAMB 2.0.5 with Cobaya, bobyqa, Planck 2018 + DESI DR2 + Pantheon+, no SH0ES). Δ is relative to the best ΛCDM fit; ΔBIC uses $N \approx 1994$ data points (estimated). The free onset and wCDM are best of 4 restarts, the others best of 8. The locked onset is a statistical tie with ΛCDM on AIC and is disfavoured on BIC; it narrowly misses the pre-registered rule $\Delta\chi^2 < -4$. Do not compare these values with those in Table 2, which use a different engine.

The results, all reported whichever way they fell, are as follows.

1. **Free onset.** $\Delta\chi^2 = -6.43$ for three extra parameters, $\Delta{\rm AIC} = -0.43$: a tie. The preferred onset $z_c = 3575$ is within 1% of matter–radiation equality for the same cosmology, $z_{\rm eq} = 3553$.

2. **Locked onset.** With z_c removed the fit is unchanged ($\Delta\chi^2 = -6.39$), so $\Delta{\rm AIC} = -2.39$. This passes the pre-registered rule. It is a modest preference (an AIC difference of about 2), and not a blind one.

	ΛCDM	Free onset	Locked onset
$100\,\omega_b$	2.240	2.261	2.261
ω_c	0.1185	0.1260	0.1260
n_s	0.9680	0.9805	0.9805
$\tau_{\rm reio}$	0.0555	0.0553	0.0553
z_c	–	3575	3553 (locked to $z_{\rm eq}$)
θ_i	–	2.789	2.789
$F/M_{\rm Pl}$	–	0.137	0.137
r_d [Mpc]	147.45	143.16	143.16
Ω_m	0.3064	0.2995	0.2995
S_8	0.815	0.824	0.824

Table 4: Best-fit parameters for the main models of Table 2.

3. **Torsion width.** Fixing $F = \sqrt{3/2}\,M_{\rm Pl}$ worsens the fit to $\Delta\chi^2 = +14.63$; the hypothesis is excluded. The best-fit width is $F \simeq 0.137\,M_{\rm Pl} \simeq \sqrt{3/2}\,M_{\rm Pl}/8.9$. A pre-registered scan of $F = \sqrt{3/2}\,M_{\rm Pl}/N$ over $N = 7$–11 (with half-integer controls) finds a minimum at $N = 8.9$ with a $\Delta\chi^2 = 1$ half-width of 1.2; the data cannot distinguish integer N, and we make no claim.

4. **Profile in $f_{\rm EDE}$.** Fixing $f_{\rm EDE}$ on a grid and refitting all other parameters gives a minimum at $f_{\rm EDE} \simeq 0.08$, with $f_{\rm EDE} = 0.12$ indistinguishable from ΛCDM and $f_{\rm EDE} = 0.15$ excluded ($\Delta\chi^2 \simeq +29$).

5. **Planck PR4.** With CamSpec PR4 and PR4 lensing in place of Planck 2018, the free model gives $\Delta\chi^2_{\rm like} = +0.09$ (no improvement) and $\Delta\chi^2 = -3.35$ when the CamSpec Gaussian foreground priors are included, with $H_0 = 70.29$ against 67.63. Neither favours the model after the parameter penalty.

6. **Curvature.** Leaving Ω_k free gives $\Omega_k = 0.0002$ in both ΛCDM and the locked model, an improvement below one unit of χ^2: no curvature is detected. The positively curved Poincaré dodecahedral space [38], which requires $\Omega_k \simeq -0.013$, is excluded by $\Delta\chi^2 = +165$, consistent with the absence of matched circles in Planck [39].

11 Pre-registered blind test: Planck + ACT DR6

The decisive test of the locked onset is on data it was not tuned to. Before running it we fixed the rule (repository file `PREREGISTRATION_ZEQ.md`): the locked model passes if $\Delta\chi^2 < -4$ against ΛCDM on the same data (equivalently $\Delta{\rm AIC} < 0$ for two extra parameters), reporting both the total and the effective χ^2. The data are the P-ACT combination recommended by the ACT collaboration: Planck 2018 with its high-ℓ range cut where ACT takes over, ACT DR6 power spectra [34] and ACT DR6 × Planck PR4 lensing [35, 36], together with DESI DR2 BAO [29] and Pantheon+ [30, 31], with no distance-ladder prior. Both models are minimised with the same code (CAMB + Cobaya) and the same optimiser settings.

Deviation from the plan, disclosed: the pre-registration named Planck PR4 and SPT-3G as possible blind data. The run used the P-ACT combination above; SPT-3G was not included. Only the ACT part of the data is new; the Planck part overlaps the data on which the lock was chosen.

Result. Status: completed 29 October 2026; **passed**. Locked onset (2 parameters): $\Delta\chi^2 = -7.23$ (total), -7.14 (effective), $\Delta{\rm AIC} = -3.23$, $H_0 = 70.43$; $p = 0.027$ for two degrees of freedom ($\approx 2.2\sigma$). Free onset (3 parameters): $\Delta\chi^2 = -7.47$, $\Delta{\rm AIC} = -1.47$, $H_0 = 70.59$.

The lock therefore costs 0.24 in χ^2 and saves a parameter. Verdict on this test: **passed** (rule $\Delta\chi^2 < -4$); overall pre-registered verdict a **tie** because of S_8 (§13).

Data (locked model, 2 parameters, vs ΛCDM)	$\Delta\chi^2$	ΔAIC	H_0	rule
Planck 2018 + DESI DR2 + Pantheon+ (same polish)	−4.70	−0.70	70.53	pass (< -4)
P-ACT + DESI DR2 + Pantheon+ (blind)	−7.23	−3.23	70.43	pass (< -4)
Planck 2018 + DESI DR2 + DES Y5 SNe	−4.55	−0.55	70.47	pass (< -2)
Planck 2018 + DESI DR2 + DES-Dovekie SNe	−5.08	−1.08	70.53	pass (< -2)
Planck 2018 + DESI DR2 + Union3 SNe	−4.91	−0.91	70.55	pass (< -2)
Planck 2018 + DESI DR2 + Pantheon+ + S_8 (DES Y3+KiDS-1000)	−2.92	+1.08	70.81	–

Table 5: Pre-registered comparisons of the locked model against ΛCDM, each re-minimised on the same data with identical settings. The S_8 line adds a Gaussian S_8 likelihood.

12 The width module: a one-parameter model

The free fits prefer $F \simeq 0.14\,M_{\rm Pl}$, close to one ninth of the torsion value $\sqrt{3/2}\,M_{\rm Pl} = 1.2247\,M_{\rm Pl}$. Before looking at ACT DR6 we wrote two tests (`PREREG_N_PACT.md`). **T1**: with $F = \sqrt{3/2}\,M_{\rm Pl}/N$ and N free, the blind data must give $N = 9 \pm 0.5$. **T2**: with $N = 9$ fixed and the onset locked, only θ_i is free, and the model must improve on ΛCDM by $\Delta\chi^2 < -2$.

T1 **failed**: P-ACT gives $N = 8.0$ (Planck 2018 with CAMB gives $N \simeq 8.5$). T2 **passed**: $\Delta\chi^2 = -6.14$, $\Delta\text{AIC} = -4.14$, $p = 0.013$ ($\approx 2.5\sigma$) on P-ACT, and $\Delta\chi^2 = -4.46$ ($\Delta\text{AIC} = -2.46$) on Planck 2018, which is not blind because the 9 was read off Planck. We have no derivation of the divisor; it is an empirical regularity until one exists.

13 Overall verdict

The overall rule (`PREREGISTRATION_VERDICT.md`) was written before the supernova swaps, the S_8 penalties and the hilltop runs were read. *Preferred* required the blind test and the fair Planck test to pass ($\Delta\chi^2 < -4$), each of three supernova sets to give $\Delta\chi^2 < -2$, and an S_8 penalty below 2 against both of two cosmic-shear values (`PREREG_S8.md`): A, DES Y3 + KiDS-1000, $S_8 = 0.790^{+0.018}_{-0.014}$ [2]; B, KiDS-Legacy, $S_8 = 0.815^{+0.016}_{-0.021}$ [3]. The first three conditions hold (Table 5). The S_8 penalty is +2.41 against A (over the limit) and +0.39 against B. The verdict is therefore a **tie**: the preference depends on which weak-lensing survey is right. Two further results are reported beside the verdict: the fit requires the field to start near the hilltop ($\theta_i \simeq 2.7$–2.9, $\Delta\chi^2 = +4$ to $+6$ for distant starts), and the pure torsion width remains excluded.

Context. The ACT DR6 extended-model analysis [1] reports no statistically significant preference for early dark energy or for other models that raise H_0. Our best-fit improvements of ≈ 2–2.5σ do not contradict that statement and do not amount to evidence; they motivate the next blind tests (§15). Best-fit values are not posteriors, and prior-volume effects in EDE posteriors are well known [8].

14 Anticipated objections

"Torsion is negligible in cosmology." For torsion sourced by fermion spin this is correct, and §2.2 quantifies it (10^{-58}–10^{-57}). The present model's torsion is sourced by $\partial_\mu\bar{\gamma}$, and its amplitude is set by $V(\phi_i)$, not by Gn^2.

"This is just EDE with a new name." At the level of the gravitational Lagrangian, yes, and §9 says so. The contribution is a derivation of the field's existence, its kinetic normalisation, its pseudoscalar nature and its late-time return to standard gravity from the Holst action. The potential is assumed.

"Ghosts, instabilities, energy conditions." ϕ is a canonical scalar with a healthy kinetic sign. Its perturbations propagate at the speed of light in its rest frame, and $\rho + p = \dot{\phi}^2 \geq 0$. Unlike the fluid profiles of earlier drafts (Appendix A), no energy-exchange sector is needed.

"Coincidence: why z_c near matter–radiation equality?" In the free fit the data place z_c within 1% of equality, and the locked model (§5) turns this into a rule with one fewer parameter. The rule is not derived: $\Lambda \approx 0.5$ eV is still put in by hand, and the lock is a hypothesis to be tested on new data (§11), not an explanation.

"The lock was chosen after looking at the data." Correct, and stated wherever the Planck 2018 result is quoted. The AIC charge of two parameters is therefore optimistic on that data set. This is why the decision rule for the next data set was fixed and hashed before the fit was run.

"Best fits are not posteriors." Correct. We report maximum-posterior points from a local optimiser with multiple starts, not MCMC credible intervals. Best-fit $\Delta\chi^2$ values are what the AIC comparison requires, but the ΛCDM optimum itself shows a near-degeneracy of order one unit of χ^2 between the likelihood and the calibration prior (§10). Differences below about one unit should not be over-interpreted. EDE posteriors are also known to suffer prior-volume effects, which is a reason to quote profile and best-fit results [8].

"The required ω_c is far from Planck." $\omega_c \simeq 0.144$ follows from geometry (§7) for any sound-horizon solution that respects BAO. In ΛCDM the Planck error is ± 0.0012, but in EDE fits ω_c is strongly degenerate with f, so that error does not apply directly. The larger ω_c also increases late-time clustering (S_8), a known cost of this model class.

"The recombination treatment is approximate." Correct. The field holds 2–6% of the energy at z_*, which shifts recombination in ways fitting formulas do not capture. The full-Boltzmann EDE analyses [8] do capture them and apply to this Lagrangian.

"Is a varying Immirzi parameter consistent with quantum gravity?" The model is a classical effective theory, and the derivation of (7) is classical [12]. The background dynamics do not depend on where the minimum ϕ_0 sits; we place it at the black-hole-entropy value $\gamma_0 \simeq 0.2375$ [40, 41] as an *input*, so that the field relaxes to the value loop quantum gravity uses today. This choice is not a prediction of the model.

15 Predictions and how the model can fail

1. **Background (shared with EDE).** A peak energy fraction $f \sim 0.1$–0.2 near $z_c \sim 3500$, $\omega_c \approx 0.14$, and $\Omega_m \approx 0.31$. If future CMB analyses push $f < 0.05$, the model cannot address the Hubble tension.

2. **Fermion coupling (distinctive).** With fermions present, the axial torsion is sourced both by $\partial_\mu \bar{\gamma}$ and by the fermion axial current J_5^μ. Eliminating torsion therefore generates a cross term $\propto \partial_\mu \bar{\gamma}\, J_5^\mu$, a derivative coupling of ϕ to fermion spin. Its coefficient is fixed by

the Holst action and is not free. Computing it, and the resulting spin-dependent effects while ϕ rolls, is the most direct route to a signature that separates this model from generic EDE. We leave it to future work.

3. **Locked onset.** On every data set the onset must coincide with matter–radiation equality for that data set's own densities. A significant preference for a different z_c on new data falsifies Eq. (13) (§11).

4. **No cosmic birefringence.** In minimal Einstein–Cartan–Holst coupling the photon does not couple to torsion, so there is no tree-level $\phi F\tilde{F}$ term. A photon coupling can only arise through the fermion loop of the derivative coupling $\partial_\mu\phi J_5^\mu$, and for a vector-like theory this is suppressed by $(\omega/m)^2$ [42]; for CMB photons and electrons this is $\lesssim (0.3\,\mathrm{eV}/511\,\mathrm{keV})^2 \sim 3\times10^{-13}$. The minimal model therefore predicts an isotropic polarisation rotation $\beta \ll 10^{-9}$ degrees, unlike photon-coupled axion EDE [43, 44]. Planck and WMAP data show a hint of $\beta = 0.34^\circ \pm 0.09^\circ$ [45, 46]. If this is confirmed and shown to arise from early dark energy, the minimal torsion origin is excluded; Simons Observatory [47] and LiteBIRD [48] can decide. This is an effective-field-theory estimate; a full one-loop calculation in an expanding background is still needed.

5. **Laboratory spin coupling.** Today the field acts on fermions as an axial background $b_0 \simeq \frac{3}{4}\dot{\bar{\gamma}}$. At the best fit ($F = 0.137\,M_{\rm Pl}$, $\theta_i = 2.79$, $z_c \simeq 3570$, $n = 3$) the oscillation amplitude decays as $a^{-3/(n+1)}$, giving $\theta \simeq 6\times10^{-3}$ and a frequency $\lesssim 10^{-33}$ eV today, hence $b_0 \sim 6\times10^{-46}$ GeV. Bounds from Lorentz-violation tests are about 10^{-31}–2×10^{-27} GeV [49]. The model is safe, and not testable in the laboratory today.

6. **Late-time limit.** The field settles at its minimum, so the model predicts no dynamical torsion today beyond the negligible residual above. Any detected late-time variation of γ would falsify this version.

16 Conclusions

Minimal Einstein–Cartan torsion cannot shorten the sound horizon: it has the wrong sign and is 56–57 orders of magnitude too weak. Promoting the Barbero–Immirzi parameter to a field changes both facts: the torsion energy becomes the positive kinetic energy of a canonical pseudoscalar, screening follows from the field's dynamics, and with one assumed potential the model is axion-like early dark energy.

Fitted with a full Boltzmann code to Planck 2018, DESI DR2 and Pantheon+, the free model ties with ΛCDM after its parameter penalty, but it places the onset at matter–radiation equality. Locking the onset there removes a parameter at no cost in fit, giving $\Delta\mathrm{AIC} = -2.4$, $H_0 = 70.6$ and $f_{\rm EDE} = 0.078$. This is a modest, post-hoc preference, not a detection; with identical optimiser settings for both models the Planck gain is $\Delta\chi^2 = -4.70$. The pure torsion width is excluded, Planck PR4 alone shows no likelihood improvement, and curvature is consistent with zero. The minimal model predicts no cosmic birefringence, which distinguishes it from photon-coupled axion EDE. On the blind, pre-registered P-ACT test the locked onset passed ($\Delta\chi^2 = -7.23$, $\approx 2.2\sigma$), and a one-parameter version with $F = \sqrt{3/2}\,M_{\rm Pl}/9$ gave $\Delta\chi^2 = -6.14$ ($\approx 2.5\sigma$), but the divisor fitted freely is 8.0, not 9, and the S_8 cost makes the pre-registered overall verdict a tie. These are hints at the 2σ level, consistent with the ACT collaboration's finding of no significant preference for early dark energy. The open problems are the origin of the eV scale, a derivation of the lock and of the divisor from the fermion sector, a full weak-lensing likelihood in place of the S_8 penalty, posterior analyses, and blind tests on SPT-3G and the Simons Observatory with rules written first.

Data and code availability

All code, fit outputs (best-fit points and their χ^2 components), pre-registration files with their SHA-256 hashes, and the scripts that recompute every number in this paper are public at `https://github.com/dannyeldridge54/AEGIS` under the AEGIS People's License (free for individuals, students, teachers and non-profit research). A one-step script (`CHECK_MY_WORK`) re-derives every number quoted here. Results that failed are kept alongside those that passed.

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A Fluid profiles and the fixed-ω_m branch

Earlier drafts used phenomenological profiles for a torsion-like fluid with $\Omega_\tau(a_c) = f/(1-f)\,[\Omega_m a_c^{-3} + \Omega_r a_c^{-4}]$: P1, $\Omega_\tau \propto 4x^6/(1+x^6)^2$, and P2, $\Omega_\tau \propto \Theta(x-1)x^{-6}$, with $x = a/a_c$. A separately conserved fluid has $w_{\rm eff} = -1 - \frac{1}{3}\,d\ln\Omega_\tau/d\ln a$. For P1 this gives $w_{\rm eff} = -1 - 2(1-x^6)/(1+x^6)$, which reaches -3 as $x \to 0$ and violates the null energy condition. The scalar model of §3 removes this problem.

Linear response at fixed (h, ω_i), $\delta\ln r_s^*/[f/(1-f)]$: P1 -0.246, P2 -0.079 at $z_c = 3540$. The response vanishes as $z_c \to z_*$. With the EC sign, $f = 0.05$ *lengthens* r_s^* by 1.35% and gives $H_0 = 62.8$.

f	h	r_s^* [Mpc]	r_d [Mpc]	$D_C(z_*)$ [Mpc]	Ω_m	$r_d h$ [Mpc]
0	0.6736	144.27	147.10	13894.4	0.3138	99.09
0.05	0.7181	142.47	145.28	13720.6	0.2761	104.33
0.10	0.7666	140.60	143.41	13541.2	0.2423	109.93
0.15	0.8195	138.68	141.46	13355.6	0.2120	115.92
0.25	0.9421	134.59	137.34	12962.1	0.1604	129.39

Table 6: P1, $z_c = 3540$, fixed $\omega_c = 0.1200$, θ_* matched. The rise in H_0 is bought by lowering Ω_m and raising $r_d h$, both excluded by BAO.

B Code

The likelihood fits (§10–§11) use AxiCLASS through Cobaya, driven by `cmb_cloud.py` and `bestfit.py` in the repository; the locked onset is implemented as a Cobaya parameter function of (ω_b, ω_c) returning $\log_{10} a_c$ from Eq. (13). The pre-registration files are `PREREGISTRATION_F.md`, `_ZEQ.md`, `_N.md`, `_OMK.md` and `_SINH.md`. The background calculations of §6–§8 come from Python scripts using NumPy and SciPy, supplied as ancillary files. `ec_calc.py` computes the minimal-EC amplitude (§2.2). `pheno.py` and `final.py` handle the fluid profiles and the fixed- and free-ω_m fits (Appendix A). `scalar_model.py` and `run_scalar.py` solve the Barbero–Immirzi field model (§8). Quadratures use `quad` in $\ln a$ (relative tolerance 10^{-9}–10^{-10}). The field equations use `solve_ivp` (DOP853, rtol $= 10^{-10}$). Root finding uses `brentq` and `fsolve`.

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